Information Theory Mn908 and MICAS

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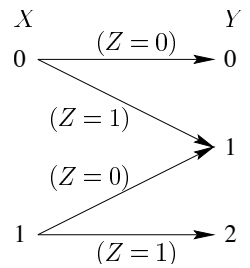
Problem 1 ***An Additive Noise Channel***

1. The channel output is $Y = X + Z$, where $X \in \{0, 1\}$ and $Z \in \{0, 45\}$. Thus, the outputs are in $\{0, 1, 45, 46\}$. If Y is 0 or 45, we know that $X = 0$. If Y is 1 or 46, we know that $X = 1$. Hence $H(X|Y) = 0$, and therefore

$$C \;=\; \max_{P_X} I(X;Y) \;=\; \max_{P_X} H(X) \;=\; 1 \text{ bit},$$

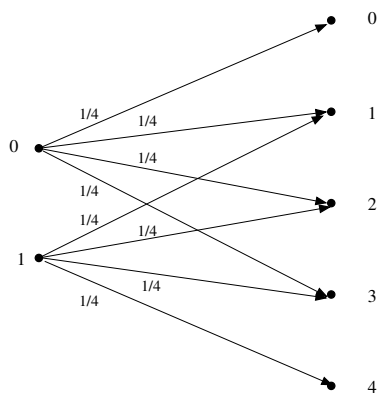
which is achieved by a uniform distribution on the input X.

2. In this case, Y has three possible output values: 0, 1, and 2. The channel looks as follows:



One sees that the channel is equivalent to a binary erasure channel with erasure probability $\alpha = \frac{1}{2}$. The capacity of the binary erasure channel is $C = 1 - \alpha = \frac{1}{2}$ bits per transmission.

3. In this case, Y has four possible values $0, 1, 2, 3, 4$ and the channel diagramme is:



Fix $P_X(1) = q$. Then,

$$H(X) = H_{\mathrm{b}}(q). \tag{1}$$

If Y is 0 or 4, there is no ambiguity on the input and thus $H(X|Y=0) = H(X|Y=4) = 0$. If Y is 1, 2, or 3 then the input can be either 0 or one. We calculate $H(X|Y=1)$. To this end, notice that by Bayes'rule and the law of total probability:

$$\begin{aligned} P_{X|Y}(1|1) &= \frac{P_{X,Y}(1,1)}{P_Y(1)} = \frac{P_{X,Y}(1,1)}{P_{X,Y}(1,1) + P_{X,Y}(0,1)} \\ &= \frac{P_X(1)P_{Y|X}(1|1)}{P_X(1)P_{Y|X}(1|1) + P_X(0)P_{Y|X}(1|1)} \\ &= \frac{q\frac{1}{4}}{q\frac{1}{4} + (1-q)\frac{1}{4}} \\ &= q. \end{aligned} \tag{2}$$

Thus,

$$H(X|Y=1) = H_{\mathrm{b}}(q). \tag{3}$$

Similarly, we also find

$$H(X|Y=2) = H(X|Y=3) = H_{\mathrm{b}}(q). \tag{4}$$

Since irrespective of the parameter q $P_Y(1) = P_Y(2) = P_Y(3) = 1/4$,

$$H(X|Y) = \frac{3}{4}H_{\mathrm{b}}(q). \tag{5}$$

By (1), we obtain

$$I(X;Y) = H_{\mathrm{b}}(q) - \frac{3}{4}H_{\mathrm{b}}(q) = \frac{1}{4}H_{\mathrm{b}}(q), \tag{6}$$

and since $H_{\mathrm{b}}(q) \leq 1$ with equality for $q = 1/2$, we deduce

$$C = \frac{1}{4}. \tag{7}$$

Problem 2 *Symbol Error Rate vs. Message Error Rate*

We will prove the following theorem:

Theorem 1. (Converse Part of the Source-Channel Coding Theorem) *Let $V_1, \ldots, V_n$ be the outcome of an i.i.d. source of law P_V. Then, if $H(V) > C$ we have*

$$\frac{1}{n}\sum_{i=1}^{n} \Pr\Big[V_i \neq \hat{V}_i\Big] \not\to 0, \tag{8}$$

i.e. it is not possible to send the process over the channel with arbitrarily low probability of bit error.

1. We have to show that Theorem 1 implies: if $H(V) > C$ then

$$\Pr\Big[(V_1, V_2, \ldots, V_n) \neq (\hat{V}_1, \hat{V}_2, \ldots, \hat{V}_n)\Big] \not\to 0.$$

To this end, it suffices to prove that:

$$\begin{aligned} \Pr\Big[(V_1, V_2, \ldots, V_n) \neq (\hat{V}_1, \hat{V}_2, \ldots, \hat{V}_n)\Big] &= \Pr\left[\bigcup_{i=1}^{n} V_i \neq \hat{V}_i\right] && (9) \\ &\geq \max_i \Pr\Big[V_i \neq \hat{V}_i\Big] && (10) \\ &\geq \frac{1}{n}\sum_{i=1}^{n} \Pr\Big[V_i \neq \hat{V}_i\Big], && (11) \end{aligned}$$

where (10) follows because the union is a larger set, and (11) follows because the maximum of a set of numbers is always larger than the average of those set of numbers. Hence, if $\frac{1}{n}\sum_{i=1}^{n} \Pr\left[V_i \neq \hat{V}_i\right]$ does not tend to zero, then $\Pr\left[(V_1, V_2, \ldots, V_n) \neq (\hat{V}_1, \hat{V}_2, \ldots, \hat{V}_n)\right]$ can neither as it is always larger.

2. We state the proof again:

$$\begin{aligned}
\frac{1}{\mathsf{K}} H\left(U_1, \ldots, U_{\mathsf{K}} \middle| Z_1, \ldots, Z_{\mathsf{K}}\right) &= \frac{1}{\mathsf{K}} \sum_{i=1}^{\mathsf{K}} H\left(U_i \middle| U_1, \ldots, U_{i-1}, Z_1, \ldots, Z_{\mathsf{K}}\right) && (12)\\
&\leq \frac{1}{\mathsf{K}} \sum_{i=1}^{\mathsf{K}} H\left(U_i \middle| Z_1, \ldots, Z_{\mathsf{K}}\right) && (13)\\
&\leq \frac{1}{\mathsf{K}} \sum_{i=1}^{\mathsf{K}} H_{\mathrm{b}}(P_{\mathrm{e},i}) + \frac{1}{\mathsf{K}} \sum_{i=1}^{\mathsf{K}} P_{\mathrm{e},i} \log(a-1) && (14)\\
&\leq H_{\mathrm{b}}\left(\frac{1}{\mathsf{K}} \sum_{i=1}^{\mathsf{K}} P_{\mathrm{e},i}\right) + \log(a-1) \frac{1}{\mathsf{K}} \sum_{i=1}^{\mathsf{K}} P_{\mathrm{e},i} && (15)\\
&= H_{\mathrm{b}}(P_{\mathrm{b}}) + P_{\mathrm{b}} \log(a-1). && (16)
\end{aligned}$$

Equation (12) follows by the chain rule, (13) follows because conditioning reduces entropy, (14) follows by Fano's inequality, and (15) follows from Jensen's inequality and because entropy is concave.

3. Let V^n be the outcomes of an i.i.d source and $\hat{V}^n$ be the reconstruction sequence. Let X^n and Y^n be the corresponding inputs and outputs of the channel. We prove the converse to the source-channel separation theorem for memoryless channels (though it holds more generally.) Let C equal the capacity of the channel.

   Then

$$\begin{aligned}
H(V) &= \frac{1}{n} H(V_1, ..., V_n) && (17)\\
&= \frac{1}{n}\Big(H(V_1, ..., V_n) + H(V_1, \ldots, V_n | \hat{V}_1, \ldots, \hat{V}_n)\\
&\quad - H(V_1, \ldots, V_n | \hat{V}_1, \ldots, \hat{V}_n)\Big) && (18)\\
&= \frac{1}{n} H(V_1, \ldots, V_n | \hat{V}_1, \ldots, \hat{V}_n) + \frac{1}{n} I(V_1, \ldots, V_n; \hat{V}_1, \ldots, \hat{V}_n) && (19)\\
&\leq \frac{1}{n} H(V_1, \ldots, V_n | \hat{V}_1, \ldots, \hat{V}_n) + \frac{1}{n} I(X_1, \ldots, X_n; Y_1, \ldots, Y_n) && (20)\\
&\leq H_{\mathrm{b}}(P_{\mathrm{b}}) + P_{\mathrm{b}} \log(|\mathcal{V}| - 1) + \frac{1}{n} I\left(X_1, \ldots, X_n; Y_1, \ldots, Y_n\right) && (21)\\
&\leq H_{\mathrm{b}}(P_{\mathrm{b}}) + P_{\mathrm{b}} \log(|\mathcal{V}| - 1) + \frac{1}{n} nC && (22)\\
&= H_{\mathrm{b}}(P_{\mathrm{b}}) + P_{\mathrm{b}} \log(|\mathcal{V}| - 1) + C && (23)
\end{aligned}$$

   where Equation (17) follows from stationarity of the source, (20) follows by the data processing inequality because $\underline{V} \to \underline{X} \to \underline{Y} \to \underline{\hat{V}}$ forms a Markov chain, (21) follows from (16) with the definition

$$P_{\mathrm{b}} = \frac{1}{n} \sum_{i=1}^{n} \Pr\left[V_i \neq \hat{V}_i\right], \qquad (24)$$

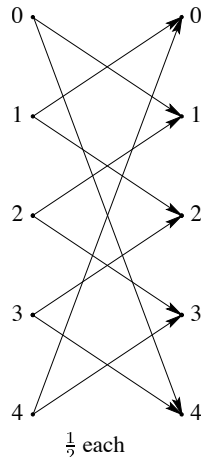
and (22) follows from the definition of capacity and because the channel is memoryless. If $P_{\mathrm{b}} \to 0$, then

$$H(V) \;\leq\; C. \tag{25}$$

We have proven now the following statement: if $P_{\mathrm{b}} \to 0$ then $H(V) \leq C$. Therefore, its contrapositive statement (which states: if $H(V) > C$ then $P_{\mathrm{b}} \not\to 0$) is also true and hence Theorem 1 is proved for memoryless channels.

Problem 3 *Zero-Error Capacity*

The channel has the following transition matrix $W(y|x)$:



$$W(y|x) \;=\; \begin{pmatrix} 0 & \frac{1}{2} & 0 & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \end{pmatrix}.$$

1. Since the channel is symmetric, its capacity is $C = \log|\mathcal{Y}| - H(\text{row of transition matrix}) = \log 5 - 1 = \log 2.5 \approx 1.322$ bits.

2. Let us construct a block code consisting of codewords of length 2. We want to achieve a capacity greater than 1 bit per channel use, i.e., greater than 2 bits per two channel uses. Thus, we need more than 4 codewords. Let us pick 5 codewords with distinct first symbols: $\{0a, 1b, 2c, 3d, 4e\}$. We must choose a, b, c, d, e so that the receiver will be able to determine which codeword was transmitted. A simple repetition code will not work, since if, say, 22 is transmitted, then 11 might be received, and the receiver could not tell whether the codeword was 00 or 22. Instead, we use the code $\{00, 13, 21, 34, 42\}$, i.e., the codewords can be described as $\mathbf{x}_i = (i, 3i \mod 5)$ for all $i = 0, \ldots, 4$. Then each codeword will be received as one of 4 possible 2-tuples which are all distinct:

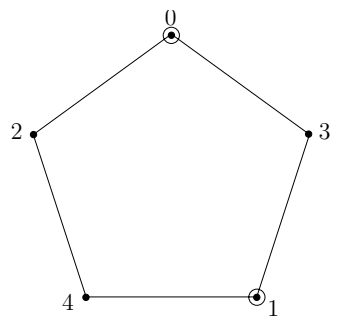
$$\begin{aligned} 00 &\to \{44, 14, 41, 11\} \\ 13 &\to \{02, 22, 04, 24\} \\ 21 &\to \{10, 30, 12, 32\} \\ 34 &\to \{23, 43, 20, 40\} \\ 42 &\to \{31, 01, 33, 03\}. \end{aligned}$$

   Since there are 5 possible error-free messages with 2 channel uses, the zero-error capacity of this channel is at least $\frac{1}{2}\log_2 5 \approx 1.161$ bits.

   In fact, the zero-error capacity of this channel is exactly $\frac{1}{2}\log_2 5$. This result was obtained by László Lovász, "On the Shannon capacity of a graph," *IEEE Transactions on Information*

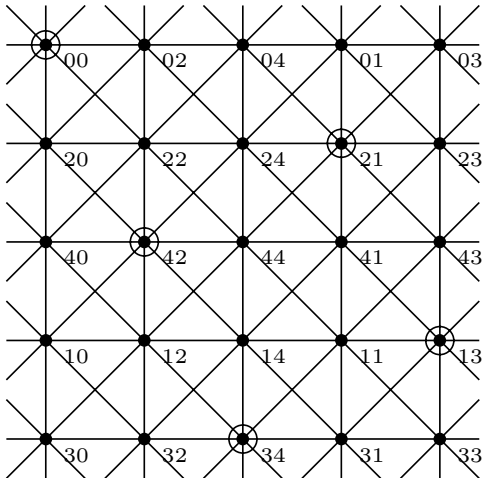
Theory, Vol IT-25, pp. 1–7, January 1979. The first results on zero-error capacity are due to Claude E. Shannon, "The zero-error capacity of a noisy channel," *IEEE Transactions on Information Theory*, Vol IT-2, pp. 8–19, September 1956, reprinted in *Key Papers in the Development of Information Theory*, David Slepian, editor, IEEE Press, 1974.

Here is an *alternative way* of solving this problem graphically. We introduce a graph consisting of vertices and undirected edges. In the case of block length $n = 1$ we introduce for each *input* symbol one vertex and two vertices are adjacent (i.e., they are connected by an edge) if and only if there is at least one *output* symbol which can be the possible output of both input symbols corresponding to the two vertices. In our example, 0 and 2 are adjacent because both can lead to the output symbol 1; but 0 and 1 are not adjacent because they have no common output symbol (see the figure below).



We see that two messages of length $n = 1$ cannot be confused if they are not adjacent, so to achieve error-freeness our codebook has to consist of non-adjacent vertices. Of course we try to take the maximum number of possible non-adjacent vertices. (In graph theory this number is called the independence number.) A choice for a codebook are e.g. the vertices 0 and 1. (By the way, because the adjacency graph for $n = 1$ for this channel looks like a pentagon, this channel is sometimes called the "pentagon channel".)

In the case of block length $n = 2$ we proceed similarly. But now, for each sequence of two input symbols we draw a vertex and there is an edge between two vertices if the two input sequences corresponding to these two vertices can be confused (i.e., if they can lead to the same output sequence). In our example, 00 and 20 are adjacent because they have the possible common output 11 (or 14). A codebook consisting of five codewords leading to error-free transmission is drawn on the next figure.



(the graph has to be understood cyclically)